

Davis Rattनाविजै

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<https://tinyurl.com/davisrattनाविजै-portfolio>

EDUCATION

Cornell University, College of Engineering, Ithaca, New York

Expected May 2028

Bachelor of Science in Mechanical Engineering, Minor in Electrical and Computer Engineering. GPA: 3.85 | Major GPA: 3.91

Courses (FA & SP): Statics and Mechanics of Materials, Dynamics, Mechanical Design, Circuits, Probability & Inference Signals

EXPERIENCE

Suspension Team Member and Purchasing Coordinator

Feb 2025 - Present

Cornell Formula SAE Racing, Cornell University, Ithaca, NY

- Collaborate with 60+ engineers to design and build a formula-style electric race car for the annual Formula SAE competition
- Designed and optimized a welding jig for all suspension links reducing jig components by 28%, size by 26%, and build time
- Work closely with suspension link designer several times a week, helping review design choices, hand calculations, and CAD
- Experience presenting at multi-stage design reviews, writing technical reports, rapid prototyping, and researching solutions
- Exposure working with mill, lathe, TIG welding, composite layups, and hand tools; basic background in PCB design
- Actively learning vehicle dynamics and system level design from literature, MATLAB scripts, and senior team members
- Oversee the purchasing team, coordinating approximately 1,200 orders annually across all engineering sub-teams

Seasonal Teller

May 2025 - Aug 2025

Middlesex Savings Bank, Wayland, MA

- Worked closely with managers, bankers, and tellers during fast-paced 44-hour weeks to run the bank's busiest branch
- Delivered customer service in-branch and over the phone to hundreds of customers (including veterans and businesses)
- Processed thousands of customer transactions, including deposits, withdrawals, loan payments, check cashing, and CDs
- Ensured day-to-day compliance with identity verification, dual control, audits, fraud detection, and documentation procedure
- Certified in financial regulations (e.g. KYC, BSA, AML); maintained strong knowledge of bank policies, products, fees

Project Lead

Sep 2024 - Jan 2025

Cornell Data Journalism, Cornell University, Ithaca, NY

- Led a team of 5 analysts to research, analyze data, and publish 3,000-word article on urging insolvency of Social Security
- Co-authored data-driven analysis on Social Security fund inflows and outflows using federal demographic and economic data
- Conducted quantitative analysis and data visualization of federal demographic and economic datasets using Python

Team Lead

Nov 2021 - May 2024

Mathematical Modeling Team, Newton North High School, Newton, MA

- Led a four-person team to three finalist placements and a 3rd place \$10k scholarship win at seven international competitions
- Co-authored seven 25-page math modeling papers; led to invitation to 2023 Joint Mathematics Meeting (JMM) in Boston
- Worked with diverse data, including categorical, time-series, and geospatial data; cleaning, modeling, and analyzing data
- Built models with linear and non-linear regressions, time-series forecasting, and multivariable calculus and linear algebra
- Used Python, MATLAB, R, and Excel on all math models for model validation, statistical analysis, and data visualization

PROJECTS

- Spatiotemporal Analysis of Lake Mead's Water Levels | Python, Data Analysis, NASA GLDAS Data, Symbolic Regression
- Quantification of Shadows in Cities with Image Analysis | MATLAB, Image Analysis, Google Earth, Non-linear Regression
- Formula SAE Car Optimized Welding Jig for Suspension Links | Autodesk Inventor (CAD), Engineering Design, Mill, Lathe
- Six Feet Structural Cardboard Model of Dragon Boat | PTC Onshape (CAD), Rapid Prototyping, 3D-Printing, Hand Tools

SKILLS

General: Teamwork, Problem Solving, Communication, Customer Service, Data Analysis, Math Modeling, Machine Shop

Software: Autodesk Inventor (CAD), Microsoft Word, Microsoft Excel, MATLAB, Simulink, Python

Languages: English, Mandarin (Conversational), Japanese (Limited Proficiency)

INTERESTS

Intramural Soccer, Mahjong, Dragon Boat Steering, Feynman Lectures, Vehicle Dynamics, Cardboard Model Building